EquipFlow • Phase 2 Requirements Engineering • D5T3

FINAL • Requirements Baseline v1.1 (ITI-Aligned)

ITI Technical Assessment | Domain D5 — Industrial Field Maintenance | Mandatory Twist T3 — Cost Governor
Source of Truth: Phase 1 — Project Definition — FINAL, Baseline v1.0
Company: EquipTech Manufacturing | **Product:** AI-powered Equipment Maintenance Assistant

This document translates the approved Phase 1 baseline and the mandatory D5T3 assignment constraints into a lean, testable MVP requirements baseline. It deliberately avoids architecture and technology decisions, except where the ITI brief mandates specific engineering constraints (e.g., Provider Abstraction, Observability, Operational Surface).

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1. Document Purpose

This document defines what EquipFlow must do and the conditions it must satisfy for the D5T3 MVP. It is the requirements baseline that will drive use cases, acceptance tests, architecture, implementation planning, security testing, evaluation, and the later gap analysis.

Phase 1 remains the authoritative product-definition baseline. Where the ITI assignment imposes mandatory D5/T3 constraints or engineering disciplines not fully specified in Phase 1, those constraints are explicitly reconciled and added below.

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2. Baseline Reconciliation (v1.0 to v1.1)

Topic	Phase 1 / v1.0 Baseline	Phase 2 / v1.1 (ITI-Aligned) Resolution
Agent count	One Maintenance Analysis Agent concept.	ITI D5 requires ≥3 specialised agents + orchestrator. MVP: Symptom

		Matcher, Diagnostic & Safety Planner, Work Order Generator.
Approval Gate	D5 requires Supervisor approval before dispatch.	Dispatch means authorising the proposed work order. Gate explicitly supports **approve / reject / edit-and-approve**, all audited.
Write Tool	CreateMaintenanceReport (Write + authorization).	The mandatory write/side-effecting tool (e.g., Dispatch Authorization) is **strictly gated** by the Supervisor approval boundary.
Cost	Not defined in Phase 1.	T3 requires per-user budgets, budget-aware routing, pre-flight estimation, hard cut-off, and spend view.
Safety	General safety mentioned.	D5 makes safety-prerequisite enforcement a **structural workflow condition**: unresolved mandatory prerequisites block dispatch authorization.
Surface & API	Preliminary API surface.	Requires a **minimal working UI/CLI** and **OpenAPI** documented HTTP API.
Observability	Basic AI execution tracing.	Requires **Correlation ID** flowing through all layers and explicit **LLM tracing**.
Engineering	Technology choices deferred.	ITI mandates **Provider Abstraction** (≥2 implementations), **versioned prompts**, and **Docker/Seed** operational requirements.

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3. MVP Scope

The MVP shall demonstrate one strong end-to-end industrial maintenance scenario while satisfying the assignment's mandatory cross-cutting requirements.

- Authentication and server-side authorization with the defined roles.

- Equipment catalog for approximately 12 equipment instances across 3 production lines.
- PDF and DOCX ingestion for the 30+ document corpus.
- Grounded RAG with hybrid retrieval, metadata filtering, citations, and refusal.
- D5 multi-step workflow using three specialised agents and an orchestrator.
- **Structural** safety-prerequisite enforcement and Supervisor approval (including edit-and-approve) before dispatch authorization.
- T3 Cost Governor: per-user budget, pre-flight estimation, budget-aware routing, hard cut-off, and spend view.
- **Minimal operational surface (UI/CLI)** and **OpenAPI** documentation.
- Full observability (Correlation IDs, LLM tracing) and evaluation harness.

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4. Actors, Responsibilities, and RBAC

Actor	Responsibility in MVP
Maintenance Technician	Ask questions, inspect permitted equipment/history, initiate analysis, review outputs.
Maintenance Engineer	Technician capabilities plus deeper analysis and review of maintenance outputs.
Maintenance Manager	Oversight, permitted document management, fleet/history visibility, trace/cost visibility.
Supervisor	Reviews proposed work orders and explicitly **approves, rejects, or edits-and-approves** dispatch authorization.
Application/Backend	Authoritative enforcement point for auth, validation, business rules, safety gating, and cost controls.
LLM Provider	Provides model capabilities only; **never** authoritative for facts, permissions, safety, or budget.

RBAC Matrix (Server-Side Enforced):

Capability	Technician	Engineer	Manager	Supervisor
Ask grounded question				
Start D5 workflow				

Ingest/Manage documents				
Approve/Edit Dispatch				
View own spend/trace				
View all spend/traces				

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5. Functional Requirements

Authentication & Authorization

- **FR-001 — Authentication:** Authenticate users before protected operations.
- **FR-002 — Role-Based Authorization:** Enforce permissions server-side. UI hiding alone is never sufficient.
- **FR-003 — Object-Level Authorization:** Verify access to requested equipment, documents, runs, and cost information.

Equipment & History

- **FR-004 — Equipment Catalog:** Provide authorised access to the MVP equipment catalog.
- **FR-005 — Equipment Identity:** Validate equipment identifiers before use in analysis.
- **FR-006 — Maintenance History:** Retrieve authorised maintenance history for an equipment instance.

Ingestion & Chunking

- **FR-007 — Document Ingestion & Chunking Strategy:** Ingest PDF and DOCX through separable stages. **The chunking strategy shall be a deliberate, documented decision** justified against the document structure (manuals, SOPs, etc.).
- **FR-008 — Document Metadata:** Preserve ID, title, type, equipment, line, version, dates, and format.
- **FR-009 — Chunk Traceability:** Each chunk shall be traceable to its source document and location (page/section).
- **FR-010 — Idempotent Re-ingestion:** Re-ingesting the same logical document/version shall not create unintended duplicates.

RAG & Retrieval

- **FR-011 — Documentation Questions:** Accept natural-language maintenance questions.
- **FR-012 — Hybrid Retrieval:** Use dense/semantic and keyword retrieval with a documented fusion method.
- **FR-013 — Metadata Filtering:** Use metadata constraints (equipment, line, version) when relevant.
- **FR-014 — Grounded Answer:** Generate answers only from evidence supplied to the generation step.
- **FR-015 — Citations:** Contain structured citations traceable to exact source chunks.
- **FR-016 — Refusal:** Refuse or request clarification when evidence is insufficient.
- **FR-017 — Conflicting Sources:** Do not silently hide material conflicts between sources/revvisions.

Tools & Write Gate

- **FR-018 — Operational Tools:** Expose controlled capabilities for equipment lookup, history, and knowledge retrieval.
- **FR-019 — Approval-Gated Write Capability:** Provide a consequential write/side-effecting capability (e.g., Dispatch Authorization). **This capability shall not execute unless all mandatory safety prerequisites are resolved AND explicit Supervisor approval exists.**
- **FR-020 — Combined Query:** Support questions requiring both documentation evidence and operational data.

Workflow, Safety & Approval

- **FR-021 — Maintenance Workflow:** Support the D5 sequence: symptom equipment/revision diagnostics safety work order.
- **FR-022 — Symptom Matching:** Identify relevant symptom/equipment evidence.
- **FR-023 — Diagnostic & Safety Planning:** Produce evidence-based diagnostics and explicit safety prerequisites.
- **FR-024 — Work Order Generation:** Produce a proposed work order from validated findings.
- **FR-025 — Work Order Lifecycle:** Represent proposed, pending approval, approved, rejected, and dispatch-authorized states.
- **FR-026 — Safety Gate:** A work order with an unresolved mandatory safety prerequisite **shall not** become dispatch-authorized (structurally enforced in code).
- **FR-027 — Supervisor Approval:** Dispatch authorization requires explicit Supervisor approval.

- **FR-028 — Approval Audit:** Record actor, action, timestamp, work-order identity, run identity, and outcome.
- **FR-029 — Edit-and-Approve:** The approval gate shall allow the Supervisor to **edit** permitted work-order fields and then approve. Any edit produces an auditable change/version and cannot bypass unresolved safety prerequisites.

Surface & API

- **FR-030 — Minimal Operational Surface:** Provide a minimal web UI or CLI/TUI covering: ingest, ask with citations, run workflow, act on approval gate, view trace, and view spend.
- **FR-031 — OpenAPI Documentation:** The HTTP API shall be documented using OpenAPI and testable from the generated specification.

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6. Agent Requirements

- **AG-001 — Minimum Agent Set:** Symptom Matcher, Diagnostic & Safety Planner, Work Order Generator, plus orchestrator.
- **AG-002 — Explicit Contracts:** Explicit input/output contracts and termination conditions.
- **AG-003 — Restricted Tools:** Explicit tool allow-list per agent.
- **AG-004 — Typed Communication:** Structured/typed contracts rather than free-form text.
- **AG-005 — No Direct Data Access:** Data access occurs through controlled application capabilities.
- **AG-006 — No Authorization Authority:** Agents do not decide user authorization, approval, or budget authority.
- **AG-007 — Evidence-Carrying Findings:** Facts carry evidence references or are marked unsupported.
- **AG-008 — Iteration Limit:** Orchestrator enforces a maximum agent iteration count.
- **AG-009 — Timeout and Retry:** Bounded timeouts and bounded retry behavior.
- **AG-010 — Graceful Degradation:** Degrade to simpler valid outcome (e.g., plain RAG) if agentic execution fails.

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7. Tool Requirements

- **TL-001 to TL-004:** Equipment Lookup, Maintenance History, Maintenance Report Search, Knowledge Retrieval (Read-only).
- **TL-005 — Write Capability:** At least one validated side-effecting operation (gated by FR-019).
- **TL-006 — Tool Validation:** Validate every tool input against its defined contract.
- **TL-007 — Tool Authorization:** Check auth, permission, agent allow-list, and business rules.
- **TL-008 — Traceability:** Record tool invocation, outcome, and association with the run/agent step.

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8. T3 — Cost Governor Requirements

- **CG-001 — Per-User Budget:** Maintain an applicable AI usage budget for each user.
- **CG-002 — Pre-Flight Estimation:** Estimate expected usage/cost before a costly workflow starts.
- **CG-003 — Budget-Aware Routing:** Select a lower-cost model/path for requests where the cheaper capability satisfies the request.
- **CG-004 — Hard Cut-Off:** Stop further billable model execution when the budget limit is reached.
- **CG-005 — Concurrency Protection:** Prevent concurrent requests from silently overspending.
- **CG-006 — Actual Usage Accounting:** Persist actual model usage and cost information.
- **CG-007 — Agent/Step Attribution:** Attribute usage to the relevant workflow/agent step.
- **CG-008 — Spend View:** Authorised users can view usage, budget, remaining budget, and run history.
- **CG-009 — Budget Failure Behavior:** If estimate exceeds budget, use cheaper path, defer, or refuse. Never silently execute.
- **CG-010 — No Bypass:** All model execution passes through the Cost Governor.

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9. Security Requirements

- **SEC-001 to SEC-011:** (Server-Side Access Control, Object Ownership, Injection Defense, Indirect Prompt Injection, Tool Output Safety, Excessive Agency, Unbounded Consumption, Sensitive Disclosure, Security Logging, Dependency/Secret Hygiene, CORS/Headers).

- **SEC-012 — Full History Secret Scanning:** A secret scanner shall be executed over the **full Git history** before submission. The repository history shall contain no secrets.

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10. Data Requirements

- **DATA-001 — Core Entities:** User, Production Line, Equipment, Maintenance Event, Maintenance Report, Document, Document Chunk, AI Conversation, AI Message, AI Execution.
- **DATA-002 — Workflow Data:** Work Order, Work Order Version, Approval Action, Safety Prerequisite, Safety Validation.
- **DATA-003 — Evidence Data:** Retrieved evidence traceable to the relevant run (Citations).
- **DATA-004 — Usage Data:** Budget, Usage Record, Model Pricing (for Cost Governor).
- **DATA-005 — Version Data:** Document revision/version information.
- **DATA-006 — Lifecycle Integrity:** State transitions for work orders and approvals validated by application rules.

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11. Non-Functional & Observability Requirements

- **NFR-001 — Groundedness:** Prefer evidence-grounded answers and correct refusal.
- **NFR-002 — Traceability:** Every AI run inspectable by run ID.
- **NFR-003 — Streaming:** Token-level streaming and live agent-progress events.
- **NFR-004 — Cancellation:** Client cancellation stops applicable server-side work.
- **NFR-005 — Reliability:** Bounded retry or explicit degradation on transient failures.
- **NFR-006 — Testability:** Testable without requiring live LLM calls.
- **NFR-007 — Provider Independence:** Business logic independent of specific LLM/Vector SDKs.
- **NFR-008 — OpenAPI:** HTTP API documented via OpenAPI.
- **NFR-009 — Session History:** Retrieve permitted conversation history.
- **NFR-010 — Health/Readiness:** Expose health/readiness endpoints.
- **OBS-001 — Correlation ID:** Each protected request receives a correlation identifier propagated to the orchestrator, agents, tools, model calls, and persisted usage records.
- **OBS-002 — LLM Tracing:** Each model call is traceable to the relevant run, agent step, model identifier, token usage, and cost.

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12. Engineering & Operational Requirements (ITI Mandated)

- **ENG-001 — AI Provider Abstraction:** Expose model capabilities through an abstraction covering completion, streaming, tool invocation, and embeddings. Swapping the provider requires configuration and one adapter only.
- **ENG-002 — Multiple Provider Implementations:** Support at least two model-provider implementations (e.g., hosted API + local/alternative) selectable by configuration, with a documented fallback chain.
- **ENG-003 — Versioned Prompt Artifacts:** Prompts shall be maintained as versioned artifacts outside business logic (not hardcoded string literals).
- **OPS-001 — Reproducible Runtime:** The MVP shall be runnable through `docker compose up` (or equivalent), bringing up the application and required data stores.
- **OPS-002 — Seed and Ingestion:** Provide a documented seed/ingest command to load the evaluation corpus and demo data.

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13. Evaluation Requirements

- **EVAL-001 — Golden Set:** ≥25 Q/A evaluation cases.
- **EVAL-002 — Adversarial Coverage:** ≥5 adversarial cases (out-of-corpus, ambiguity, injection, conflicting sources, insufficient evidence).
- **EVAL-003 — Injection Resistance:** ≥3 prompt-injection cases, including indirect injection in an ingested document.
- **EVAL-004 to EVAL-006:** Retrieval Metrics, Groundedness, Refusal Correctness.
- **EVAL-007 — Safety Gate:** Prove unresolved safety prerequisites block dispatch.
- **EVAL-008 — Cost Governor:** Cover sufficient/insufficient budget, routing, hard cut-off, concurrency.
- **EVAL-009 — Agent Controls:** Verify agent selection, tool allow-lists, termination, degradation.
- **EVAL-010 — Real Baseline:** Record actual numbers, including failures.
- **EVAL-011 — Mandatory Negative Test Matrix:** The evaluation and integration tests must explicitly cover a negative test matrix (e.g., LLM requests dispatch without approval Blocked; Supervisor approves with unresolved safety Blocked; Budget exhausted Hard cut-off).

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14. Core Use Cases

- **UC-01 to UC-04:** Grounded Question, Operational Question, Combined Question, D5 Maintenance Analysis.
- **UC-05 — Supervisor Approval (incl. Edit):** Supervisor reviews proposed work order inspects findings/safety **optionally edits fields** approves/rejects persists action only approved result becomes dispatch-authorized.
- **UC-06 to UC-08:** Budget-Limited Request, Insufficient Evidence, Prompt Injection in Document.

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15. Business Rules

- **BR-001 — Backend Authority:** App is responsible for auth, validation, safety, and cost.
- **BR-002 — Evidence First:** LLM reasons over evidence; not the authoritative source of facts.
- **BR-003 — No Evidence, No Fabrication:** Insufficient evidence requires refusal.
- **BR-004 — Revision Awareness:** Revision-sensitive guidance must use applicable revision or expose conflict.
- **BR-005 — Safety Before Dispatch:** Unresolved mandatory safety prerequisite blocks dispatch (structurally enforced).
- **BR-006 — Human Before Dispatch:** Supervisor approval is mandatory; no implicit approval.
- **BR-007 — Tool Mediation:** Model requests tool; app decides execution.
- **BR-008 — Cost Before Execution:** Budget checks occur before and during execution.
- **BR-009 — No Physical Control:** EquipFlow does not control industrial equipment.
- **BR-010 — MVP Boundary:** Not a CMMS, ERP, IoT, or scheduling product.

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16. Traceability Matrix (ITI Brief Alignment)

ITI Core Requirement	EquipFlow Phase 2 Coverage
FR-1 Ingestion	FR-007 to FR-010, ENG-003
FR-2 Retrieval	FR-012 to FR-017
FR-3 Evaluation	EVAL-001 to EVAL-011
FR-4 Multi-agent	AG-001 to AG-010, TL-001 to TL-008
FR-5 Orchestration	AG-008 to AG-010, FR-025 to FR-029 (incl. Edit-and-Approve)

FR-6 Real-time	NFR-003, NFR-004
FR-7 Surface	FR-030 (Minimal UI), FR-031 (OpenAPI)
FR-8 Access	FR-001 to FR-003, RBAC Matrix
FR-9 Observability	OBS-001, OBS-002, NFR-002, CG-006, CG-007
Architecture	ENG-001, ENG-002, ENG-003, NFR-007
Packaging	OPS-001, OPS-002
Security	SEC-001 to SEC-012
T3 Cost Governor	CG-001 to CG-010, EVAL-008

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17. Phase 2 Exit Criteria

- All mandatory MVP capabilities have unique requirement IDs.
- D5 safety enforcement is structural and testable (FR-026).
- Supervisor approval explicitly includes **edit-and-approve** (FR-029).
- T3 Cost Governor controls are explicit and testable.
- ITI engineering mandates (Provider Abstraction, OpenAPI, Minimal UI, Docker/Seed, Observability) are explicitly captured.
- Every critical workflow can be translated into acceptance tests (including the Negative Test Matrix).
- No requirement requires a technology choice to understand the required behavior.

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18. Requirements Decisions Deferred to Architecture/Design

- LLM provider/model selection and fallback implementation (covered by ENG-001/002).
- Embedding model selection.
- Vector store/search technology.
- Relational database technology and exact schema.
- Agent framework/orchestration library.
- API framework and UI technology (covered by FR-030/031).
- Streaming transport implementation.
- Deployment/cloud infrastructure.

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19. Final MVP Definition

EquipFlow D5T3 MVP is a controlled industrial-maintenance copilot that can ingest and retrieve maintenance knowledge, answer grounded questions with traceable citations, access authorised operational data through tools, execute a bounded three-agent diagnostic/safety/work-order workflow, **structurally prevent dispatch when safety prerequisites are unresolved**, require explicit **Supervisor approval (including edit-and-approve)** before dispatch authorization, govern AI consumption through the **T3 Cost Governor**, and provide a **minimal operational surface** with full **observability**, all built on a **provider-agnostic, cleanly architected** foundation.

Document control: Phase 2 — Requirements Engineering — FINAL / Requirements Baseline v1.1 (ITI-Aligned).
